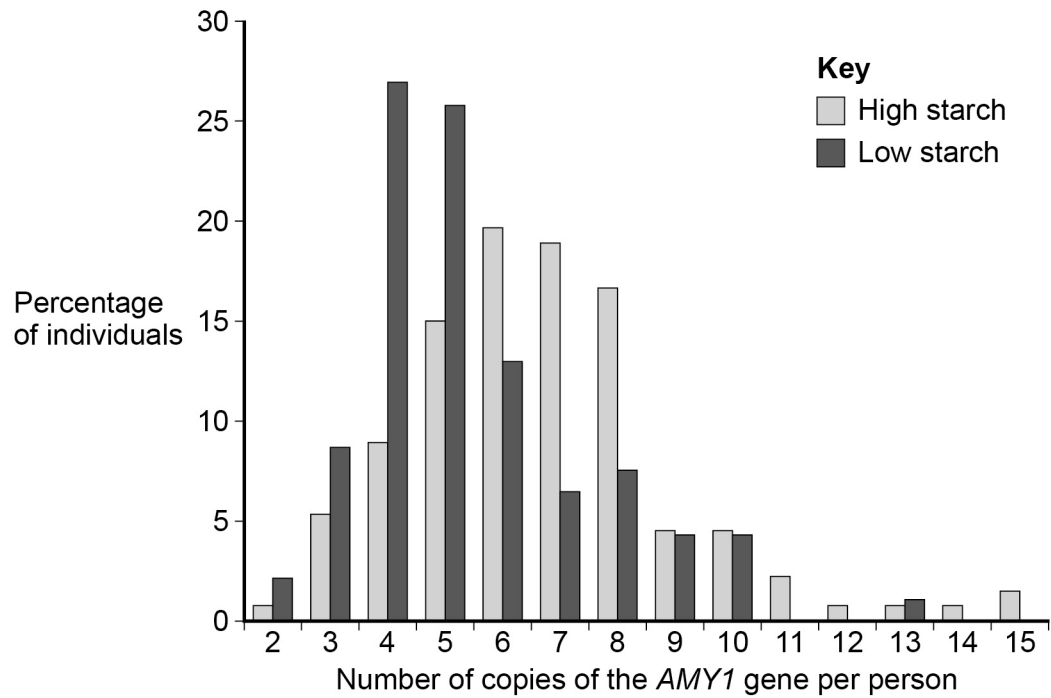


0 3 . 1

Most human cells contain two copies of each gene. However, there might be up to 15 copies of the gene for amylase (*AMY1*). Scientists investigated the number of copies of the *AMY1* gene in individual people in two populations. One population had a high-starch diet and the other population had a low-starch diet.

Figure 3 shows their results.

Figure 3



Describe what their results show.

[3 marks]



[Extra space]

0 3 . 2

Multiple copies of the *AMY1* gene is an adaptation to a high-starch diet.

Use your knowledge of protein synthesis and enzyme action to explain the advantage of this adaptation.

[3 marks]

[Extra space]

Question 3 continues on the next page

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Multiple copies of the *AMY1* gene is an adaptation to a high-starch diet.

Suggest how this evolved through natural selection.

[3 marks]

[Extra space]

9

